Ahmad Mousavi

Department of Mathematics and Statistics
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https://scholar.google.com/citations?user=IStw0S4AAAAJ&hl=en

Positions

- **American University** Washington, DC
  Assistant Professor of Data Science, Department of Mathematics and Statistics Fall 2024–Present

  Affiliate Faculty, Institute of Artificial Intelligence, Kogod School of Business 2024–Present

  Faculty Fellow, Center for Data Science, School of Public Affairs 2026–Present

  Professorial Lecturer, Department of Mathematics and Statistics Fall 2023–Spring 2024
- **University of Florida** Gainesville, FL
  Postdoctoral Associate, Informatics Institute 2021-2023
- **University of Minnesota** Minneapolis, MN
  Industrial Postdoctoral Fellow, Institute for Mathematics and its Applications 2019-2021

Research Summary

I am an Assistant Professor of Data Science in the Department of Mathematics and Statistics at American University. My recent work concerns large language models, multimodal learning, and NLP applications in social media and healthcare. Earlier and ongoing work includes sparse optimization, kernel-free quadratic classifiers, penalty decomposition methods, mathematical finance, and decentralized bilevel programming.

Google Scholar: https://scholar.google.com/citations?user=IStw0S4AAAAJ&hl=en (311 citations; h-index 8).

Research Interests

- General: Natural Language Processing, Machine Learning, Deep Learning, Sparse Optimization, Large Scale Optimization, Mathematical Finance, Mathematical Modeling, Quantum Computing, and Numerical Linear Algebra.
- Specific (not restricted to): Support Vector Machines, Penalty Decomposition Method, ℓ_1-Minimization, Greedy Algorithms, Semidefinite Programming, ADMM, Inexact Newton's Methods, Matrix Decomposition, Linear Programming, Quadratic Optimization, and so on.

Education

- **University of Maryland, Baltimore County** Baltimore, MD

Ph.D. in Applied Mathematics 2013-2019

- Advisor: Prof. Jinglai Shen
- Thesis topic: Topics in Sparse Recovery via Constrained Optimization: Least Sparsity, Solution Uniqueness, and Constrained Exact Recovery

- **University of Maryland, Baltimore County** Baltimore, MD
  M.Sc. in Applied Mathematics 2013-2015
- **Sharif University of Technology** Tehran, Iran
  M.Sc. in Applied Mathematics 2008-2011
  - Advisor: Prof. Nazam Mahdavi-Amiri
  - Thesis topic: An Inexact Newton Method for Nonconvex Equality Constrained Optimization
- **University of Guilan** Rasht, Iran
  B.Sc. in Applied Mathematics 2004-2008

Selected Papers

Journal Articles

17. Zohre Aminifard, Saman Babaie-Kafaki, Zeinab Saeidian, and **Ahmad Mousavi**. "A Dynamically Nonmonotone Trust Region Algorithm Based on Scalar Estimations of the Hessian with Application to Binary Classification with Outliers." *Journal of Computational and Applied Mathematics*, Article 117998, 2026. DOI: 10.1016/j.cam.2026.117998. `https://doi.org/10.1016/j.cam.2026.117998`.

16. **Ahmad Mousavi**, Yeganeh Abdollahinejad, Roberto Corizzo, Nathalie Japkowicz, and Zois Boukouvalas. "E-CaTCH: Event-Centric Cross-Modal Attention with Temporal Consistency and Class-Imbalance Handling for Misinformation Detection." *IEEE Transactions on Computational Social Systems*, pp. 1–14, 2026. DOI: 10.1109/TCSS.2026.3693647. `https://doi.org/10.1109/TCSS.2026.3693647`.

15. Hossein Moosaei, Milan Hladík, **Ahmad Mousavi**, Zheming Gao, and Haojie Fu. "Quadratic Surface Twin Support Vector Machine for Imbalanced Data." *International Journal of Machine Learning and Cybernetics* 17, no. 5 (2026): 250. `https://link.springer.com/article/10.1007/s13042-026-03059-8`.

14. Yeganeh Abdollahinejad, Sayed Mohsin Reza, B. Ashwini, and **Ahmad Mousavi**. "Systematic Literature Review of Machine Learning Methods for Emotion Recognition Using EEG and Physiological Signals in Healthcare." *IEEE Access* (2026). DOI: 10.1109/ACCESS.2026.3681521. `https://doi.org/10.1109/ACCESS.2026.3681521`.

13. Yeganeh Abdollahinejad, **Ahmad Mousavi**, Petros Siaplaouras, Zois Boukouvalas, and Roberto Corizzo. "EMO-CARE: EEG Multi-Scale Temporal Modeling with Channel-Aware Feature Attention for Robust Subject-Independent Emotion Recognition." *IEEE Open Journal of the Computer Society* (2026). `https://ieeexplore.ieee.org/document/11348088`.

12. **Ahmad Mousavi**, Maziar Salahi, and Zois Boukouvalas. "Sparse Extended Mean-Variance-CVaR Portfolios with Short-Selling." *Japan Journal of Industrial and Applied Mathematics* 43, no. 1 (2026): 12. `https://link.springer.com/article/10.1007/s13160-025-00760-z`.

11. Parvin Nazari, **Ahmad Mousavi**, Davoud Ataee Tarzanagh, and George Michailidis. "A Penalty-Based Method for Communication-Efficient Decentralized Bilevel Programming." *Automatica* 173 (2025): 112039. `https://doi.org/10.1016/j.automatica.2024.112039`.

10. **Ahmad Mousavi** and George Michailidis. "Cardinality Constrained Mean-Variance Portfolios: A Penalty Decomposition Algorithm." *Computational Optimization and Applications* (2025). `https://doi.org/10.1007/s10589-025-00653-4`.

9. **Ahmad Mousavi** and George Michailidis. "Statistical Proxy Based Mean-Reverting Portfolios with Sparsity and Volatility Constraints." *International Transactions in Operational Research* 32, no. 6 (2025): 3848–3869. `https://doi.org/10.1111/itor.13442`.

8. Hassan Rezapour, Ramin Nasiri, and **Ahmad Mousavi**. "The Hyper-Zagreb Index of Trees and Unicyclic Graphs." *Iranian Journal of Mathematical Sciences and Informatics* 18, no. 1 (2023): 41–54. `https://ijmsi.ir/browse.php?a_id=1356`.

7. Hossein Moosaei, **Ahmad Mousavi**, Milan Hladík, and Zheming Gao. "Sparse ℓ_1-Norm Quadratic Surface Support Vector Machine with Universum Data." *Soft Computing* 27, no. 9 (2023): 5567–5586. `https://doi.org/10.1007/s00500-023-07860-3`.

6. **Ahmad Mousavi** and Jinglai Shen. "A Penalty Decomposition Algorithm with Greedy Improvement for Mean-Reverting Portfolios with Sparsity and Volatility Constraints." *International Transactions in Operational Research* 30, no. 5 (2023): 2415–2435. `https://doi.org/10.1111/itor.13123`.

5. **Ahmad Mousavi**, Zheming Gao, Lanshan Han, and Alvin Lim. "Quadratic Surface Support Vector Machine with ℓ_1 Norm Regularization." *Journal of Industrial & Management Optimization* 18, no. 3 (2022): 1835–1861. `https://doi.org/10.3934/jimo.2021046`.

4. Sai K. Popuri, Nagaraj K. Neerchal, Amita Mehta, and **Ahmad Mousavi**. "Density Estimation Using Entropy Maximization for Semi-Continuous Data." *Digital Signal Processing* 116 (2021): 103107. `https://doi.org/10.1016/j.dsp.2021.103107`.

3. **Ahmad Mousavi**, Mehdi Rezaee, and Ramin Ayanzadeh. "A Survey on Compressive Sensing: Classical Results and Recent Advancements." *Journal of Mathematical Modeling* 8, no. 3 (2020): 309–344. `https://jmm.guilan.ac.ir/article_4155.html`.

2. **Ahmad Mousavi** and Jinglai Shen. "Solution Uniqueness of Convex Piecewise Affine Functions Based Optimization with Applications to Constrained ℓ_1 Minimization." *ESAIM: Control, Optimisation and Calculus of Variations* 25 (2019): 56. `https://doi.org/10.1051/cocv/2019007`.

1. Jinglai Shen and **Ahmad Mousavi**. "Least Sparsity of p-Norm Based Optimization Problems with $p > 1$." *SIAM Journal on Optimization* 28, no. 3 (2018): 2721–2751. `https://doi.org/10.1137/17M1140066`.

Conference Papers

1. Javad Rajabi, Sunday Okechukwu, **Ahmad Mousavi**, Roberto Corizzo, Charles C. Cavalcante, and Zois Boukouvalas. "Event-Based Multi-Modal Fusion for Online Misinformation Detection in High-Impact Events." Proceedings of the IEEE International Conference on Big Data, pp. 3301–3308, 2024. `https://ieeexplore.ieee.org/abstract/document/10826054`

Submitted and Non-Peer-Reviewed Manuscripts

13. Mojtaba Soltanalian and **Ahmad Mousavi**. "The Rank-Collapse Principle for Quadratic Optimization." arXiv preprint arXiv:2608.07828, 2026. `https://arxiv.org/abs/2608.07828`

12. Aruthra Satish Kumar, Yeganeh Abdollahinejad, Shahriar Khosravi, Majid Alikhani, Hadi Talebighadikolahi, and **Ahmad Mousavi**. "CAM-Soft: Scalable Multimodal Fusion for Soft Hate Speech Detection via Camouflaged-Implicature Modeling and Counterfactual Invariance." Manuscript in preparation (intended for IEEE BigData).

11. Kanishka Jayathilake, **Ahmad Mousavi**, and Ramin Ayanzadeh. "Weaver: A Hybrid Quantum-Classical Optimization Method for Binary Compressive Sensing under Matrix Uncertainty." Submitted to AAAI 2027 (July 2026).

10. Miguel Esparza, Aydin Ayanzadeh, **Ahmad Mousavi**, and Ali Mostafavi. "Physics-Conditioned Attention Bias and Retrieval-Augmented Correction for Next-Day Wildfire Spread Prediction." Preprint, July 2026.

9. Ping Yang, Jian Luo, **Ahmad Mousavi**, Roberto Corizzo, and Zheming Gao. "An Ensemble Learning Approach Based on Random Sparse Quadratic Twin Parametric-Margin SVMs." Available at SSRN 7022004 (2026). Submitted to *Knowledge-Based Systems* (June 2026). `https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7022004`

8. Md Toufikuzzaman, **Ahmad Mousavi**, and Dongwon Lee. "BLADE: Bilevel Low-Rank Augmented-Lagrangian Erasure for LLM Unlearning." Submitted to EMNLP 2026 via ACL ARR 2026 May (Submission 8792).

7. Yeganeh Abdollahinejad, **Ahmad Mousavi**, Naeemul Hassan, Kai Shu, Nathalie Japkowicz, Shahriar Khosravi, and Amir Karami. "MOMENTA: Mixture-of-Experts over Multimodal Embeddings with Neural Temporal Aggregation for Misinformation Detection." arXiv preprint arXiv:2604.16172 (2026). Under review at *Information Fusion*. `https://arxiv.org/abs/2604.16172`

6. **Ahmad Mousavi** and Majid Alikhani (equal contribution), Yeon-Chang Lee, Roberto Corizzo, and Yeganeh Abdollahinejad. "MURAL: Multimodal Uncertainty-Aware Recommendation via Adaptive Edge Learning." In revision for conference submission (earlier version declined at RecSys 2026).

5. **Ahmad Mousavi**, Morteza Kimiaei, Saman Babaie-Kafaki, and Vyacheslav Kungurtsev. "An Efficient Penalty Decomposition Algorithm for Minimization over Sparse Symmetric Sets." arXiv preprint arXiv:2601.12383, 2026 (under review at *Mathematical Programming Computation*). `https://arxiv.org/abs/2601.12383`

4. **Ahmad Mousavi**, Ramin Zandvakili, and Zheming Gao. "ℓ_0-Regularized Quadratic Surface Support Vector Machines." arXiv preprint arXiv:2501.11268, 2025. `https://arxiv.org/abs/2501.11268`

3. Ramin Ayanzadeh, **Ahmad Mousavi**, Amirhossein Basareh, Narges Alavisamani, and Kazem Taram. "Enigma: Application-Layer Privacy for Quantum Optimization on Untrusted Computers." arXiv preprint arXiv:2311.13546, 2023. `https://arxiv.org/abs/2311.13546`

2. Jinglai Shen and **Ahmad Mousavi**. "Exact Support and Vector Recovery of Constrained Sparse Vectors via Constrained Matching Pursuit." arXiv preprint arXiv:1903.07236, 2019. `https://arxiv.org/abs/1903.07236`

1. Ramin Ayanzadeh, **Ahmad Mousavi**, Milton Halem, and Tim Finin. "Quantum Annealing Based Binary Compressive Sensing with Matrix Uncertainty." arXiv preprint arXiv:1901.00088, 2019. `https://arxiv.org/abs/1901.00088`

Books

- Hassan Rezapour and **Ahmad Mousavi**. Probability and Statistics (A Comprehensive Book for M.S. Nationwide Examination in Economics and Management Fields, Sobhan-e Mehr Publication, 2013, In Farsi).

Honors and Awards

- CAS Faculty Grant for Scholarship and Creative Inquiry, College of Arts and Sciences, American University, $2,500 ("Fair Collaborative Learning via Synthetic Payloads in Imbalanced Settings"), awarded December 2025.
- Associate Postdoctoral Fellowship, Informatics Institute, University of Florida.
- Industrial Postdoctoral Fellowship, Institute for Mathematics and its Applications (IMA), University of Minnesota.
- Outstanding Graduate Research Award in Mathematics, College of Natural and Mathematical Sciences, UMBC, 2019.
- Lodging Support to Attend Foundation of Data Science Summer School, Georgia Institute of Technology, 2019.
- Full Graduate Assistantship from Department of Mathematics and Statistics, University of Maryland, Baltimore County, 2013-2019.
- ICERM Travel and Lodging Support to Attend Optimization Methods in Computer Vision and Image Processing Workshop, Providence, Rhode Island, USA, 2019.
- ICERM Lodging Support to Attend Computational Imaging Workshop, Providence, Rhode Island, USA, 2019.
- SIAM Student Travel Award to Attend SIAM Annual Meeting, Portland, Oregon, USA, 2018.
- UMBC Graduate School Professional Development Grant to Attend SIAM Annual Meeting, 2018, Portland, Oregon, USA.
- UMBC Graduate School Professional Development Grant to Attend Optimization Methods in Computer Vision and Image Processing Workshop, Providence, Rhode Island, USA, 2019.
- Ranked 13th in Nationwide Entrance Examination for Ph.D. Program in Mathematics, Iran, April 2011.
- Ranked 16th among 10763 Participants in the Nationwide Entrance Examination for M.S. Program in Mathematics, Iran, February 2008.
- Ranked 4th among Fellow M.S. Students in Applied Mathematics, Department of Mathematical Sciences, Sharif University of Technology.
- Ranked 1st among Fellow B.S. Students, Faculty of Mathematical Sciences, The University of Guilan.

- Accepted with Full Reimbursement for Autumn School Algorithmic Optimization, Trier University, 2016 (However, I could not attend).

Work Experience

- Research and Development Intern, Precima, R&D division, Chicago, IL, Summer 2019.

Teaching Experience

- DATA 441/641: Applied Natural Language Processing, Spring 2025.
- DATA 442/642: Advanced Machine Learning, Fall 2024; Fall 2025 (two sections).
- STAT/DATA 427/627: Statistical Machine Learning, Spring 2024, Fall 2024, Spring 2025, Summer 2025.
- DATA 412/612: Statistical Programming in R, Fall 2023, Spring 2024.
- MATH 225: Introduction to Differential Equations, Summer and Winter 2018, and Winter 2019.

Supervising Experience

- **Data Science Practicum:**
  - *Fall 2024:* Yen-Chun Lin, Sean Hsu, Ting Yi Chuang, and Po-Yu Lai.
    Fairness in Imbalanced Data — Guided students in exploring Universum-based quadratic and linear SVM models to address bias in imbalanced datasets, with applications such as fraud and disease detection.
  - *Spring 2025:* Meet Patel, Dwanith Venkat Girish.
    Neural Network Pruning — Supervised the development and evaluation of structured and unstructured pruning methods to improve the efficiency and scalability of deep learning models.
  - *Fall 2025:* Peter Ozo-Ogueji, Kaitlin Works-Figueroa, Aidan Hennessy, and Kennya Jiles.
    Multimodal Emergency Department Diagnosis Discordance — Led students in designing multimodal models integrating clinical notes, vitals/labs time series, and medical imaging (MIMIC-IV) to predict diagnostic discordance between ED admissions and discharge diagnoses.
- **Independent Study / Capstone:**
  - *Fall 2024:* Spencer Grewe (DATA 690).
    Medical Computer Vision — Supervised architectural and empirical analysis of U-Net for biomedical image segmentation, leading to development of LFASNet.
- **External and Cross-Institutional Supervision:**
  - *Spring 2026–Present:* Yanjia Zhang, Pooja Jitendra Shah, and Sai Praneet Reddy Chinthala (George Mason University), co-supervised with Dr. Lindi Liao (George Mason University); Katherine Lessard (American University; Spring 2026, discontinued before completion).
    Fair Bilevel Optimization with Synthetic Data for Collaborative Classification — Guiding students in developing and implementing an efficient bilevel optimization algorithm for fairness-aware collaborative learning using synthetic data generation.

- *2026–Present:* Aruthra Satish Kumar (George Mason University).
  CAM-Soft: Scalable Multimodal Fusion for Soft Hate Speech Detection via Camouflaged-Implicature Modeling and Counterfactual Invariance — Mentoring research on this manuscript (in preparation; intended for IEEE BigData).
- *Spring 2025–Present:* Yeganeh Abdollahinejad (Michigan State University, Department of Biology; mentorship began while she was at Pennsylvania State University).
  Robust Multi-Modal and Multi-Scale Learning under Distribution Shift — Mentoring work on "EMO-CARE: EEG Multi-Scale Temporal Modeling with Channel-Aware Feature Attention for Robust Subject-Independent Emotion Recognition," "E-CaTCH: Event-Centric Cross-Modal Attention with Temporal Consistency and Class-Imbalance Handling for Misinformation Detection," "MOMENTA: Mixture-of-Experts over Multimodal Embeddings with Neural Temporal Aggregation for Misinformation Detection" (under review at *Information Fusion*), and "CAM-Soft: Scalable Multimodal Fusion for Soft Hate Speech Detection via Camouflaged-Implicature Modeling and Counterfactual Invariance" (manuscript in preparation; intended for IEEE BigData), including model design, experiments, and manuscript preparation.
- *Fall 2025–Present:* Md Toufikuzzaman (Pennsylvania State University), co-advised with Dr. Dongwon Lee (Pennsylvania State University).
  BLADE: Bilevel Low-Rank Augmented-Lagrangian Erasure for LLM Unlearning — Co-advising the student in the constrained bilevel unlearning framework (submitted to EMNLP 2026), including theoretical formulation, implementation, and evaluation on TOFU, MUSE, and KnowUndo.
- *2025–Present:* Majid Alikhani (independent researcher, Toronto).
  Multimodal Recommendation and Soft Hate Speech Detection — Supervised research on "MURAL: Multimodal Uncertainty-Aware Recommendation via Adaptive Edge Learning" (in revision after RecSys 2026) and "CAM-Soft: Scalable Multimodal Fusion for Soft Hate Speech Detection via Camouflaged-Implicature Modeling and Counterfactual Invariance" (manuscript in preparation; intended for IEEE BigData).
- *2026–Present:* Miguel Esparza (Texas A&M University) and Aydin Ayanzadeh (UMBC), co-mentored with Dr. Ali Mostafavi (Texas A&M University).
  Physics-Conditioned Attention Bias and Retrieval-Augmented Correction for Next-Day Wildfire Spread Prediction — Co-mentoring research on this preprint (July 2026).

Review Experience

- Review for the following journals and conferences: Neural Networks, NeurIPS 2025 Workshop on Constrained Optimization in Machine Learning (COML), IEEE Transactions on Signal Processing, Journal of Optimization Theory and Applications, Mathematical Methods of Operations Research, Optimization Methods and Software, Digital Signal Processing, Physica A: Statistical Mechanics and its Applications, Numerical Algorithms, Set-Valued and Variational Analysis, Mathematical Problems in Engineering, Infor: Information Systems and Operational Research, Complex and Intelligent Systems, and so on.

Conference/Workshops Presentation and Attendance

- (Poster Presentation) E-CaTCH: Event-Centric Cross-Modal Attention with Temporal Consistency and Class-Imbalance Handling for Misinformation Detection, AU Research Conference, American University, Washington, DC, USA, May 11, 2026.

- (Attendance) StatConnect@AI, Georgetown University, Washington, DC, USA, 2026.
- (Attendance) NeurIPS 2025, San Diego, CA, USA, 2025.
- (Attendance) IEEE International Conference on Data Mining (ICDM 2025), Washington, DC, USA, 2025.
- (Presentation) Exploring Sparsity: Modeling and Algorithmic Insights in Machine Learning and Finance, Departmental Seminar, Department of Mathematics, Semnan University, October 13, 2025.
- (Attendance) Workshop on Intersections between Control, Learning, and Optimization, UCLA, Los Angeles, CA, USA, 2019.
- (Poster Presentation) Solution Uniqueness of Convex Piecewise Affine Functions Based Optimization with Applications to Constrained ℓ_1 Minimization, Princeton Day of Optimization, 2018; ICERM Computational Imaging Workshop, 2019; ICERM Optimization Methods in Computer Vision and Image Processing Workshop, 2019.
- (Presentation) Some Topics in Sparse Optimization, `http://www.norbertwiener.umd.edu/seminars/`, The Norbert Wiener Center, University of Maryland, College Park, MD, USA, 2018.
- (Presentation) Solution Uniqueness of Convex Piecewise Affine Functions Based Optimization with Applications to Constrained ℓ_1 Minimization, `http://meetings.siam.org/sess/dsp_programsess.cfm?SESSIONCODE=65264`, SIAM Annual Meeting, Portland, OR, USA, 2018.
- (Attendance) SIAM Annual Meeting, Pittsburgh, PA, USA, 2017.
- (Presentation) A Mathematical Introduction to Compressive Sensing, University of Maryland, Baltimore County, Optimization Seminar, Baltimore, MD, USA, 2016.
- (Attendance) American Mathematical Society Spring Eastern Sectional Meeting, Baltimore, MD, USA, 2014.
- (Attendance) 3rd and 4th Workshop on Optimization and its Applications, Tehran, Iran, 2011 and 2012.
- (Attendance) 3rd and 4th International Conference of Iranian Operations Research Society, Tehran and Rasht, Iran, May 2010 and 2011.
- (Attendance) 40th Annual Iranian Mathematics Conference, Tehran, Iran, 2009.

Service

- Member, AI Steering Committee, American University, Fall 2025 – Present.
  - I also serve as a member of both the Student Engagement and Scholarship Group and the Responsible AI Group within the committee.
- Member, Student Research Committee, Department of Mathematics and Statistics, American University, Fall 2025 – Present.
- Program Committee Member, 5th IEEE Big Data Workshop on Multimodal AI (MMAI 2025), IEEE International Conference on Big Data, 2025.
- Program Committee Member, Multimodal AI (MMAI) Workshop, IEEE International Conference on Big Data, 2026.

- External PhD Thesis Reviewer and Defense Examiner, Mohammadigohari Mahdi, Computer Science, Free University of Bolzano (38th Cycle).
- Judge, *Smith Analytics Consortium Datathon*, Robert H. Smith School of Business, University of Maryland (UMD-American University student analytics competition sponsored by Deloitte), 2025 and 2026.
- Vice President, Mathematics and Statistics Graduate Student Association (MSGSA), University of Maryland, Baltimore County, 2015.
- Orientation Advisor, University of Maryland, Baltimore County, Summer 2017.

Miscellaneous

- **Computer Skills**
  - R
  - Python
  - MATLAB
  - LaTeX
  - Microsoft Office
- **Memberships**
  - IEEE (Regular Member)
  - Society for Industrial and Applied Mathematics (SIAM; previous)